Log replay and benchmark toolchain

Nov 8, 2026 [name to be added] about 90 seconds

One-minute script

> This slide is how we test changes without driving again. Every run writes a log. A decoder turns it into a table, a narrator prints what the rover was thinking at each moment, and a Python copy of the firmware rules replays the log. So a change can be tried on old runs before we flash it.
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> Two more tools sit on top. The landmark benchmark measures the same landmarks in nine logs and keeps them in one file, so we know which numbers are normal. The simulators add people, weather and slip to real scans.
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> Here is what it did this week. The chart shows where the replay turns on seven runs. The old rule turned too early in log 8, at 18.2 metres; the new one turns at 21.8. On the others it stays where the good runs turned. If we shift the start of a run by up to 8 metres, the turn moves by at most 0.13 metres.
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> It also found the cause of the three failures: the follower in log 8, the start-dependent window in campus 9, and the hold that was released too early. The limit: it is a copy of the firmware, so it has to be kept in sync, and distances are commanded, not measured, because we have no wheel encoders.

What you need to understand

The pipeline: log file (.bin from the box) → decoder (table) → narrate_run (plain-language story of the run) → landmark_benchmark (numbers per log) → fwmirror (Python copy of the rules) → replay_turn / sweep_start / simulators → results → firmware → flash.

- Why a copy: the real firmware runs on the ESP32. The copy runs the same rules on the laptop in seconds, on every old log, so we see the effect of a change before it can break the rover.
- Start-shift sweep: we cut or pad the start of each log by −8 to +8 m. A good rule turns at the same place; ours moved by 0.00 to 0.13 m.
- Benchmark: nine logs, same functions as the firmware. It showed, for example, the pole-line distance 6.5 to 6.8 m in five runs, and a B wall behind within ±1.3 degrees in six.
- We do not tune to bad runs: a drifted run is used to find a cause, not to set a number.

Likely questions

Does the replay match the real rover? It uses the same rules and logged sensor data, but it is a Python copy. A change in the C++ code has to be copied over; we check the numbers against the logs.

Why not just drive again? Every outdoor run costs time and risk to the rover. Replay tells us in seconds which change is worth driving.

What is not covered? Real crowds, real weather and wheel slip are simulated, not measured. Distances are commanded, not from encoders.